

Your new dataset looks like a **content/document approval and marketing metadata feed**.

This type of dataset is extremely suitable for:

- data quality anomaly detection
- business rule anomaly detection
- metadata drift detection
- content governance monitoring
- regulatory/compliance monitoring

Your dataset has:

- document lifecycle fields
- approval dates
- expiry dates
- product mappings
- country/platform/language metadata
- marketing classifications
- delivery information

This means anomaly detection becomes very important because bad data here can cause:

- compliance issues
- expired content usage
- wrong product promotions
- invalid campaign targeting
- broken links

- duplicate documents
 - downstream reporting failures
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1. What is Anomaly?

An anomaly is:

Data that deviates from expected normal behavior.

Simple example from your dataset:

Field	Normal	Anomaly
expiry_date	after approved_start_date	before approved_start_date
document_status	Approved/Expired	random unexpected value
RowCount	50k/day	suddenly 2k
country	valid countries	NULL spike
link	proper URL	malformed URL
product_apims_id	mapped correctly	duplicates/missing
language	EN/FR/DE	sudden unknown values

2. Main Types of Anomalies

There are 3 major anomaly types.

A. Point Anomaly

Single abnormal record.

Example:

document_id	subjectline length
D1	45
D2	60
D3	10000 ← anomaly

Another example:

expiry_date < approved_start_date

B. Contextual Anomaly

Abnormal only under certain context.

Example:

- Japanese language content in US-only campaigns
- Certain products appearing in wrong country

Example:

country	language	Status
France	FR	Normal
France	EN	Maybe anomaly

Context matters:

- country
 - product
 - platform
 - content type
 - delivery source
-

C. Collective Anomaly

Group behavior becomes abnormal.

Example:

- 40% documents suddenly missing product_apims_id
- huge spike in expired documents
- entire feed losing link column values

This usually indicates:

- ETL issue
 - upstream API issue
 - schema issue
-

3. Best Anomaly Categories For Your Dataset

Your dataset is best suited for these anomaly categories.

A. Volume Anomalies

Detect:

- row drops
- row spikes
- missing feeds

Metrics:

- RowCount
- Sum

Example:

- Daily documents usually = 100k
- Today = 8k

→ anomaly

B. Completeness Anomalies

Detect NULL spikes.

Critical columns:

- document_id
- approved_start_date
- country
- product
- link
- document_status

Metric:

- Completeness

Example:

- product becomes 60% NULL suddenly
-

C. Uniqueness Anomalies

Detect duplicate records.

Critical:

- document_id
- approved_job_id
- based_on_id

Metrics:

- Uniqueness
- Distinct ValuesCount
- Unique ValueRatio

Example:

```
SELECT document_id, COUNT(*)  
FROM table  
GROUP BY document_id  
HAVING COUNT(*) > 1
```

D. Validity Anomalies

Check invalid formats or business logic.

Examples:

Invalid Dates

expiry_date < approved_start_date

Invalid URLs

link NOT LIKE 'http%'

Invalid Language Codes

language NOT IN ('EN','FR','DE','JP')

Metric:

- ColumnValues
 - CustomSql
-

E. Distribution Anomalies

Detect shifts in data patterns.

Examples:

- sudden increase in one country
- product distribution changes
- unusual language spikes

Metrics:

- Entropy
- Mean
- StandardDeviation

Example:

- US documents normally 40%
- suddenly 95%

→ anomaly

F. Schema Anomalies

Detect:

- column missing
- datatype changes
- length anomalies

Metrics:

- ColumnCount
- ColumnLength

Example:

- subjectline suddenly contains massive HTML blobs
-

G. Relationship Anomalies

Detect broken relationships.

Examples:

Relationship	Validation
country ↔ language	expected mapping
product ↔ indication	valid association
classification ↔ content_type	allowed combination

Metric:

- ColumnCorrelation

4. Best Implementation Approaches

For enterprise feeds like yours:

LEVEL 1 — Rule-Based Detection (BEST START)

This is the most important.

Most companies rely heavily on this.

Examples

Completeness Check

```
SELECT COUNT(*)  
FROM table  
WHERE document_id IS NULL
```

Date Validation

```
SELECT *  
FROM table  
WHERE expiry_date < approved_start_date
```

Duplicate Detection

```
SELECT document_id, COUNT(*)  
FROM table
```

```
GROUP BY document_id
HAVING COUNT(*) > 1
```

URL Validation

```
WHERE link NOT LIKE 'http%'
```

Advantages:

- simple
- explainable
- stable
- enterprise-friendly

LEVEL 2 — Statistical Anomaly Detection

Detect deviations from historical trends.

Example:

Day	Row Count
Mon	100k
Tue	102k
Wed	98k
Thu	5k ← anomaly

Using Z-score:

```
genui  {"math_block_widget_always_prefetch_v2":{"content":"z = \frac{x-\mu}{\sigma}"}}
```

If:

- $z > 3$
→ anomaly

Best for:

- RowCount
- NULL %
- country distributions
- document volume trends

LEVEL 3 — ML-Based Detection

Useful later when:

- rules become too many
- patterns become complex

Best algorithms for your data:

Algorithm	Use
Isolation Forest	best overall
LOF	local anomalies
Autoencoder	advanced deep learning
DBSCAN	clustering anomalies

Use ML for:

- hidden pattern anomalies
- multidimensional anomalies
- drift detection

5. Best Metrics From Your Available List

Your metrics are actually excellent.

HIGH PRIORITY

Metric	Use
Completeness	NULL spikes
RowCount	feed health
Uniqueness	duplicate detection
Distinct ValuesCount	cardinality drift
ColumnValues	invalid values
CustomSql	business rules

MEDIUM PRIORITY

Metric	Use
Entropy	distribution drift
StandardDeviation	statistical anomalies
Unique ValueRatio	duplicate spikes
ColumnCorrelation	relationship validation

LOWER PRIORITY

Metric	Use
Mean	numeric drift
Sum	aggregate anomalies
ColumnLength	corrupted text
ColumnCount	schema changes

6. Critical Business Rules For Your Dataset

These are extremely valuable.

Date Rules

Expiry Validation

`expiry_date >= approved_start_date`

Document Status Rules

Expired documents should not have ACTIVE status

Product Mapping Rules

`product_apims_id` should map correctly to product

Country-Language Rules

Examples:

Country	Expected Language
France	FR
Germany	DE
Japan	JP

URL Rules

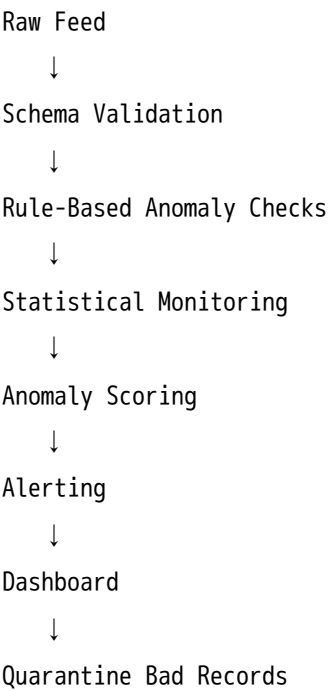
All links should start with https

Content Classification Rules

Email content must have subjectline

7. Enterprise Architecture Recommendation

Recommended pipeline:



8. Best Tools For Your Use Case

Strongly recommended:

Tool	Why
Great Expectations	easiest enterprise validation
AWS Deequ	excellent for Spark
Soda	modern observability

Tool	Why
dbt tests	warehouse checks
Airflow	orchestration
Datadog	monitoring

9. Recommended Real-World Implementation

Phase 1 (START HERE)

Implement:

- RowCount
- Completeness
- Duplicate detection
- Date validations
- Enum validations
- URL checks

This gives huge value immediately.

Phase 2

Add:

- historical baselines
 - statistical thresholds
 - entropy drift
-

Phase 3

Add ML:

- Isolation Forest
 - clustering anomalies
 - behavioral drift detection
-

10. Recommended Anomaly Checks For Your Dataset

CRITICAL CHECKS

Feed-Level

- RowCount anomaly
 - schema change
 - missing partition detection
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Document-Level

document_id

- uniqueness
- completeness

approved_start_date

- valid dates

expiry_date

- business validation

link

- URL validation

language

- accepted values

product

- mapping validation
-

Business-Level

Expired document usage

expired documents should not remain active

Missing subjectline

email content without subjectline

Country mismatch

country-language inconsistency

11. Most Important Enterprise Insight

Real enterprise systems usually use:

Approach	Usage
Rule-based	70%
Statistical	20%
ML	10%

Reason:

- easier to explain
- easier to audit
- more stable
- easier to maintain

12. Final Recommendation For Your Dataset

START WITH THESE

Priority order:

1. Completeness
 2. RowCount
 3. Uniqueness
 4. ColumnValues
 5. CustomSql
 6. Distinct ValuesCount
 7. Entropy
 8. ColumnCorrelation
-

BEST FIRST IMPLEMENTATION

Implement:

- NULL checks
- duplicate checks
- invalid dates
- invalid URLs
- invalid enum values
- historical row count comparison
- country-language validation
- expired document validation

This will already provide enterprise-grade anomaly detection for your feeds.